

DDA5001 Machine Learning

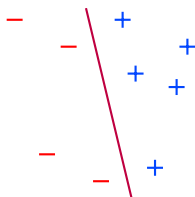
Maximum Likelihood Estimation (MLE) & MLE Interpretation of Least Squares (Week 2, Part 2)

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Recap: Linear Classification



Linear classification is to separate the training dataset:

$$\begin{cases} f_{\theta}(\mathbf{x}_i) > 0 & \text{if } y_i = +1, \\ f_{\theta}(\mathbf{x}_i) < 0 & \text{if } y_i = -1, \end{cases} \quad \forall i = 1, \dots, n.$$

The linear classification model is represented as

$$f_{\theta}(\mathbf{x}) = \theta^{\top} \mathbf{x}, \quad y = \text{sign}(f_{\theta}(\mathbf{x})).$$

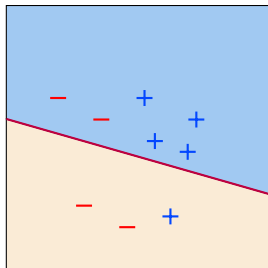
Recap: The Perceptron

- Pick a **misclassified** data $\mathbf{x}_i$ (any misclassified one)

$$\underbrace{\text{sign}(f_{\boldsymbol{\theta}}(\mathbf{x}_i))}_{=\text{sign}(\boldsymbol{\theta}^{\top} \mathbf{x}_i)} \neq y_i.$$

- Update rule

$$\boldsymbol{\theta} = \boldsymbol{\theta} + y_i \mathbf{x}_i.$$



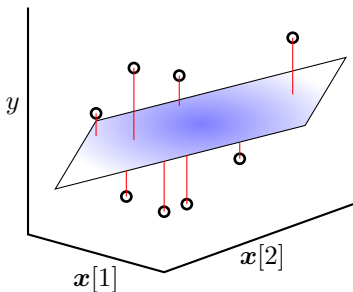
- The perceptron is an iterative algorithm.
- It may converge to **any** linear classifier that can classify the training data points.
- It is designed for **binary classification** and for **linearly separable** data.

Recap: Linear Regression

Linear regression is to use a linear model $f_{\theta}(\mathbf{x})$ to fit the **continuous real-valued** label y . Namely,

$$y_i \approx f_{\theta}(\mathbf{x}_i) = \boldsymbol{\theta}^{\top} \mathbf{x}_i,$$

where $\boldsymbol{\theta} \in \mathbb{R}^d$.



Least squares is a machine learning algorithm for finding such a linear regression model:

$$\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Recap: Solution of Least Squares

Let (we can ignore the $\frac{1}{n}$ without loss of generality)

$$\mathcal{L}(\boldsymbol{\theta}) = \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Then

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) &= (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) \\ &= \boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}\end{aligned}$$

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Facts:

- ▶ If $h(\boldsymbol{\theta}) = \mathbf{c}^\top \boldsymbol{\theta}$, then $\nabla h(\boldsymbol{\theta}) = \mathbf{c}$.
- ▶ If $h(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{M} \boldsymbol{\theta}$ ($\mathbf{M}$ is symmetric), then: $\nabla h(\boldsymbol{\theta}) = 2\mathbf{M}\boldsymbol{\theta}$.

Recap: Solution of Least Squares

Let (we can ignore the $\frac{1}{n}$ without loss of generality)

$$\mathcal{L}(\theta) = \|X\theta - y\|_2^2$$

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$$\begin{aligned}\mathcal{L}(\theta) &= (X\theta - y)^\top (X\theta - y) \\ &= \theta^\top X^\top X\theta - 2\theta^\top X^\top y + y^\top y\end{aligned}$$

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Take the gradient

$$\nabla_{\theta} \mathcal{L}(\theta) = 2X^\top (X\theta - y)$$

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Setting the gradient to zero and due to **convexity** of $\mathcal{L}$, the optimal solution $\hat{\theta}$ satisfies

$$X^\top X\hat{\theta} = X^\top y$$

How to solve for $\hat{\theta}$?

Recap: Solution of Least Squares: Case I

Case I: $\mathbf{X} \in \mathbb{R}^{n \times d}$ has full column rank

- ▶ $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}$ will be full rank / invertible (Homework 1).
- ▶ We have

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{y}.$$

Pseudo-inverse of matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ when it has full column rank is defined as

$$\mathbf{X}^\dagger = \left(\mathbf{X}^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \in \mathbb{R}^{d \times n}.$$

- ▶ We have $\mathbf{X}^\dagger \mathbf{X} = \mathbf{I}$, but not necessarily $\mathbf{X} \mathbf{X}^\dagger = \mathbf{I}$.

Hence, we can also write

$$\hat{\boldsymbol{\theta}} = \mathbf{X}^\dagger \mathbf{y}.$$

Recap: Solution of Least Squares: Case I

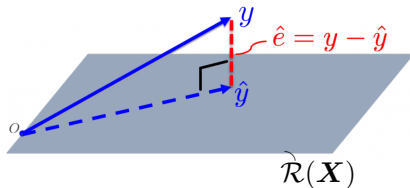
Geometric interpretation:

- Let

$$\mathcal{P}_X = \mathbf{X}\mathbf{X}^\dagger.$$

- $\mathcal{P}_X$ is the **projection matrix** onto the range space $\mathcal{R}(\mathbf{X})$.
- What does the LS solution mean?

$$\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{y} - \mathbf{X}\mathbf{X}^\dagger\mathbf{y} = \mathbf{y} - \mathcal{P}_X(\mathbf{y}) := \mathbf{y} - \hat{\mathbf{y}}.$$



Conclusion: The LS solution is such that $\hat{\mathbf{y}} := \mathbf{X}\hat{\boldsymbol{\theta}} = \mathcal{P}_X(\mathbf{y})$ is the **orthogonal projection** of $\mathbf{y}$ onto the range space $\mathcal{R}(\mathbf{X})$.

Recap: Solution of Least Squares: Case II

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We can represent all the solutions using singular value decomposition (SVD), and it will be discussed in Homework 1.

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How to obtain a unique and meaningful solution in this case?

↪ Regularization, which we will study in details later in the overfitting chapter (simple derivation will be discussed in Homework 1).

Agenda

Maximum Likelihood Estimation (MLE)

MLE Interpretation of Least Squares

Feasibility of Learning

Maximum Likelihood Estimation: A Simple Example

Flip Coin

- ▶ Setup: For a 'special' coin (with head and back), flip it.
- ▶ Outcome: Dataset $\mathcal{D} = \{H, H, B, H, B, \dots\}$, k heads out of n flips.
- ▶ Question:

What is the probability it will be head?

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Assumption:

- ▶ The outcome of every flip follows a Bernoulli distribution:
 $\Pr[\text{head}] = \theta$ and $\Pr[\text{back}] = 1 - \theta$.
- ▶ Flips are i.i.d.
 - The $i + 1$ -th flip is independent of i -th flip.
 - The number of heads follow the binomial distribution.

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Task: Learning θ from data using Maximum Likelihood Estimation (MLE).

Conditional Probability and Likelihood

- ▶ The conditional probability

$$\Pr[y|x]$$

means **the probability of y given x** . An important quantity for learning ($\rightsquigarrow$ later course on logistic regression and language modeling)

- ▶ How about

$$\Pr[\mathcal{D}|\theta]?$$

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- ▶ How about

$$\Pr[\mathcal{D}|\theta]?$$

Likelihood of data

$\Pr[\mathcal{D}|\theta]$: The probability of observed data given parameter θ .

Write down the likelihood:

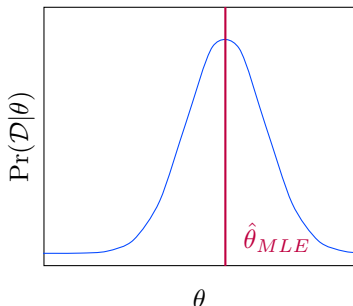
$$\Pr[\mathcal{D}|\theta] = \theta^k (1 - \theta)^{n-k}$$

since we have k heads out of n trials and i.i.d.

Maximum Likelihood Estimation

Maximum likelihood estimation: The principle

Maximize the probability of observed data over parameter θ .



Log-likelihood:

$$\log(\Pr[\mathcal{D}|\theta]) = k \log(\theta) + (n - k) \log(1 - \theta)$$

Maximum Likelihood Estimation

Set $\mathcal{L}(\theta) = \log(\Pr[\mathcal{D}|\theta])$, we have

$$\frac{\partial \mathcal{L}(\theta)}{\partial \theta} = \frac{k}{\theta} - \frac{n-k}{1-\theta}$$

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Why log-likelihood?

► $\log(\cdot)$ is increasing, thus

$$\max_{\theta} \Pr[\mathcal{D}|\theta] \iff \max_{\theta} \log(\Pr[\mathcal{D}|\theta])$$

► $\log(\Pr[\mathcal{D}|\theta])$ is often easier to maximize than $\Pr[\mathcal{D}|\theta]$.

Maximum Likelihood Estimation: General Procedure

- ▶ Observe (i.i.d.) data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ following $p(\mathbf{x}|\boldsymbol{\theta}^*)$.
- ▶ Build the likelihood function $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^n p(\mathbf{x}_i|\boldsymbol{\theta})$ for some parameter $\boldsymbol{\theta}$.
- ▶ Log-likelihood $\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \log(p(\mathbf{x}_i|\boldsymbol{\theta}))$
- ▶ Maximum likelihood estimator: $\hat{\boldsymbol{\theta}}_{MLE} = \operatorname{argmax}_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$.

Key point: Know $p(\mathbf{x}|\boldsymbol{\theta})$.

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Feasibility of Learning

Recall: Multivariate Gaussian distribution

- **Gaussian distribution.** A random variable X is said to follow $\mathcal{N}(\mu, \sigma^2)$ (Gaussian distribution with mean μ and variance σ^2) if its probability density function (PDF) is given by

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

- **Multivariate Gaussian distribution.** We say the random vector $\mathbf{X} \in \mathbb{R}^d$ follows Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (assumed to be PD), if its PDF is given by

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

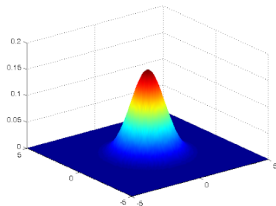
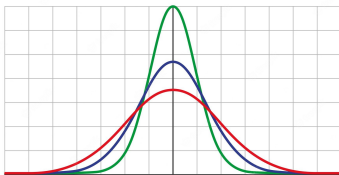
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LS for Linear Regression: MLE Interpretation

- ▶ Recap LS:

$$\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

- ▶ Recap our model:

$$y_i \approx f_{\boldsymbol{\theta}}(\mathbf{x}_i) = \boldsymbol{\theta}^\top \mathbf{x}_i.$$

- ▶ To be more explicit, consider

$$y_i = \boldsymbol{\theta}^\top \mathbf{x}_i + \epsilon_i \quad \text{assume} \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where ϵ_i are i.i.d. Gaussian for $i = 1, \dots, n$.

- ▶ In matrix notation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon} \quad \text{with} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Sigma} = \text{diag}(\sigma^2, \dots, \sigma^2)$.

LS for Linear Regression: MLE Interpretation

Equivalently,

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The likelihood: The probability of observed data given parameters $\boldsymbol{\theta}$

$$\begin{aligned} p(\mathcal{D}|\boldsymbol{\theta}) &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}\boldsymbol{\epsilon}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\epsilon}\right) \quad \leftarrow \text{PDF of Gaussian distribution} \\ &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})\right). \end{aligned}$$

Log-likelihood:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= \log(p(\mathcal{D}|\boldsymbol{\theta})) = \text{constant} - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \\ &= \text{constant} - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2. \end{aligned}$$

LS for Linear Regression: MLE Interpretation

MLE principle:

$$\hat{\boldsymbol{\theta}}_{MLE} = \operatorname{argmax}_{\boldsymbol{\theta}} -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 = \operatorname{argmin}_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2,$$

which is exactly LS.

Take-home message

Least square is a maximum likelihood estimator under Gaussian noise assumption

How Good Is This Estimator? Unbiased or Biased.

Suppose the target hypothesis g is $g(\mathbf{X}) = \mathbf{X}\boldsymbol{\theta}^*$ and

$$\mathbf{y} = g(\mathbf{X}) + \epsilon = \mathbf{X}\boldsymbol{\theta}^* + \epsilon \quad \text{with} \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Goal: To learn the target $\boldsymbol{\theta}^*$ (then we know g to the most extent).

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Definition: An estimator of a given parameter is said to be **unbiased** if its expected value is equal to the true/underlying value of the parameter.

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Unbiased Estimator:

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_{MLE}] = \boldsymbol{\theta}^*$$

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Is Learning Possible? When and How?

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- ▶ We can even have **adversarial** training and testing data, in which training data says nothing (even worse) about testing data.

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- ▶ The **(really) unknown** target function g is the object of learning. Learning is all about to infer g **outside of the seen training dataset**.
- ▶ There are extreme cases, e.g., we know g on the training dataset, but nothing on other unseen (test) data points. Learning in this case is obviously infeasible.
- ▶ Know things we have already seen, this is not learning, it is **memorizing**.

Learning is Possible in a Probabilistic Way

From the previous example, we know that learning is not feasible if we

- ▶ make no assumptions about the connections between the training and test data points, and
- ▶ want to firmly (**deterministically**) predict something about the test data.

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From the previous example, we know that learning is not feasible if we

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Fortunately, learning will be possible if we

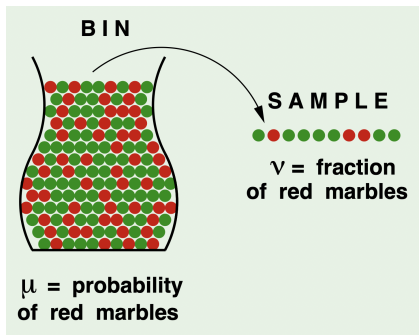
- ▶ make some assumptions that the training and testing data are related in some way. The most common assumption:

The training and test data are independent and identically distributed (i.i.d.)

- ▶ predict something about the test data **in a probabilistic way**.

A Bin Sampling Example: i.i.d. and Randomness

- ▶ Consider a bin with **red** and **green** marbles.
 - $\Pr[\text{red marble}] = \mu$.
 - $\Pr[\text{green marble}] = 1 - \mu$.
- ▶ The learning task is to learn μ .
- ▶ μ is **unknown** to us.
- ▶ We pick n marbles **randomly** and **independently** (with **replacement**). This means i.i.d.



The **fraction** of red marbles is v .

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It is “**Possible** versus **Probable**”.

- ▶ **No.** **Learning is not possible** if you want to be absolutely certain
 - ▶ There is some probability that the samples are mostly **green**, where learning fails.
- ▶ **Yes.** **Learning is possible** in a probabilistic way.
 - ▶ The sample frequency ν is close to μ with high probability.

Concentration Inequality

We can use tools from statistics to establish this “probable” learning result, i.e., **concentration inequality**. That is, to quantify “with high probability”.

We can have

$$\Pr[|\nu - \mu| > \varepsilon] \leq 2e^{-2n\varepsilon^2} \quad \forall \varepsilon > 0.$$

This result is immediately from **Hoeffding's inequality**.

- ▶ The result is valid for any n and ε .
- ▶ With proper n and ε , it says that ν is close to μ **with high probability** (i.e., probably). That is, $\nu \approx \mu$ is **probably and approximately correct (P.A.C.)**.

↪ In the sequel, we will study this training versus testing result for general machine learning such as classification and regression.

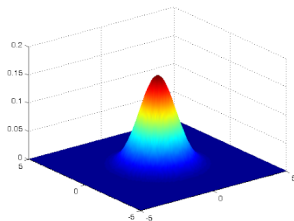
Appendix

MLE for Estimating Gaussian Parameters

MLE for Gaussian

Recall Multi-variate Gaussian distribution:

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$



Data: Draw a set of i.i.d. samples $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ follow $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$.

Qes: What is the $\hat{\boldsymbol{\theta}}_{MLE}$ for $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ given data?

MLE for Gaussian

Likelihood: Set parameters $\theta = (\mu, \Sigma)$

$$\begin{aligned} p(\mathcal{D}|\theta) &= \prod_{i=1}^n p(\mathbf{x}_i|\theta) \\ &= \frac{1}{(2\pi)^{nd/2}} \frac{1}{|\Sigma|^{n/2}} \prod_{i=1}^n \exp\left(-\frac{1}{2}(\mathbf{x}_i - \mu)^\top \Sigma^{-1}(\mathbf{x}_i - \mu)\right) \end{aligned}$$

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Log-likelihood: Set parameters $\theta = (\mu, \Sigma)$

$$\begin{aligned} \mathcal{L}(\theta) &= \sum_{i=1}^n \log(p(\mathbf{x}_i|\theta)) \\ &= -\frac{nd}{2} \log(2\pi) - \frac{n}{2} \log(|\Sigma|) - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \mu)^\top \Sigma^{-1}(\mathbf{x}_i - \mu) \end{aligned}$$

MLE for Gaussian

Optimization:

- We first optimization over μ , compute the **gradient**

$$\nabla_{\mu} \mathcal{L}(\theta) = \Sigma^{-1} \sum_{i=1}^n (x_i - \mu).$$

Setting the gradient to be 0, together with the fact that Σ^{-1} is positive definite, provides

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i.$$

MLE for Gaussian

Optimization:

- We then optimize over Σ . Note that $|\Sigma|^{-1} = |\Sigma^{-1}|$. Taking gradient over Σ^{-1} yields

$$\nabla_{\Sigma^{-1}} \mathcal{L}(\boldsymbol{\theta}) = \frac{n}{2} \Sigma - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{MLE})(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{MLE})^\top$$

setting the gradient to zero gives

$$\hat{\Sigma}_{MLE} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{MLE})(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_{MLE})^\top$$

Used facts:

$$\nabla_{\mathbf{A}} \log(|\mathbf{A}|) = \mathbf{A}^{-\top}$$

and

$$\nabla_{\mathbf{A}} \text{tr}(\mathbf{A}^\top \mathbf{B}) = \mathbf{B}.$$

Unbiased and Biased Estimators

MLE for the mean of Gaussian is **unbiased**:

$$\mathbb{E} [\hat{\boldsymbol{\mu}}_{MLE}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbf{x}_i] = \boldsymbol{\mu}^{\star}$$

MLE for the variance of Gaussian is **biased**:

$$\mathbb{E} \left[\hat{\boldsymbol{\Sigma}}_{MLE} \right] \neq \boldsymbol{\Sigma}^{\star}$$